

Overview

For this simple Chameleon project, you will be setting up a digital soil prong powered by a BBC micro:bit. The goal of this project is to grow an assortment of microgreens and lentils. The soil prong will provide you when to water the plants you set up & will locally collect moisture, light, and temperature data for Team Chameleon to study. Once the project is over, you can then upload the file onto a Google Form which we will use to optimize our curriculum and environmental understanding.

Seed Prep

Inside the tote bag you received, there will be a small parchment paper pouch, tied with twine. Untie this pouch with caution, since there are many small seeds inside. The seeds are an assortment of lentils, and before they can be planted in dirt, they need to be soaked overnight. Place the seeds in a container of water and let it sit for 8 hours. After this, the seeds are ready to be planted.

Soil Prep

For the soil and pot setup, anything will suffice. As long as your pot is big enough to hold the soil prong, it'll support the seeds too. Soil that is most like garden dirt is optimal, and dirt with wood chips or stones should try to be avoided.

Micro:bit Setup

1. take out battery pack and insert two batteries inside
2. unfold cardstock battery sleeve and follow folding steps specified on the front
3. once the battery sleeve is on, take out the micro:bit and the kritonik soil moisture prong
4. grab three screws and three bolts from the bag
5. insert the screws into the corresponding holes on the micro:bit and the prong labeled P1, 3V and GND
6. Once these are secured, use the empty holes on the cardstock sleeve to secure the microbit to the battery pack
7. Grab the dangling wire from the battery pack and insert it into the large white port on the left of the micro:bit
8. Write code at makecode.microbit.org/#editor*

9. Here is the code for the simple version:

```
reading = 0
```

```
def on_forever():  
    global reading  
    reading = pins.analog_read_pin(AnalogPin.P0)  
    led.plot_bar_graph(reading, 1023)  
    if input.button_is_pressed(Button.A):  
        basic.show_number(reading)  
basic.forever(on_forever)
```

10. Download code off the makeCode platform

11. Insert your soil moisture sensor into a houseplant or some samples you gather and watch the magic happen!

(all readings from 0-1023 are in microvolts (mV) signifying the conductivity of your soil)

***Micro:bit Code**

No coding or applications need to be installed in order to program your micro:bit. Instead, microsoft makecode will be used and all data collected by your micro:bit setup will be downloaded locally.